11.	Implement an application that writes data to the SD card.

AIM:

To develop an Android application using Android Studio that allows users to enter data and store it in the device's SD card (external storage).

ALGORITHM:

1. Input:

- The user enters text data in the application.
- The user clicks a button to save the data into the SD card.

2. Steps for Implementation in Android Studio:

1. Create a New Android Project:

- Open **Android Studio**
- Create a new project with an Empty Activity
- Choose Java or Kotlin as the programming language

2. Design the User Interface:

- Add the following components in the layout:
  - EditText → to enter data
  - Button → to save data
  - TextView (optional) → to display status messages

3. Add Permissions in Manifest File:

- Declare permission to write to external storage
- For modern Android versions, also handle runtime permissions

4. Handle Runtime Permissions:

- Check if permission is granted
- If not, request permission from the user at runtime
- Proceed only after permission is granted

5. Check External Storage Availability:

- Ensure that:
    - SD card is available
    - SD card is writable

6. Create File in SD Card:

- Define a file name (e.g., mydata.txt)
- Select appropriate directory:
    - App-specific folder (recommended)
    - Or public directory

7. Write Data to SD Card:

- Read input from EditText
- Open file output stream
- Write the data into the file

- close the stream

8. Display Output to User:

- Show confirmation message:
  - "Data saved successfully"
  - Or error message if something fails

9. Handle Exceptions:

- Manage:
  - IOException
  - Permission denial
  - Storage unavailability

3. Testing the Application:

- Run the app on a real device/emulator
- Enter sample text
- Click Save button
- Verify file using file manager in external storage

4. Observations:

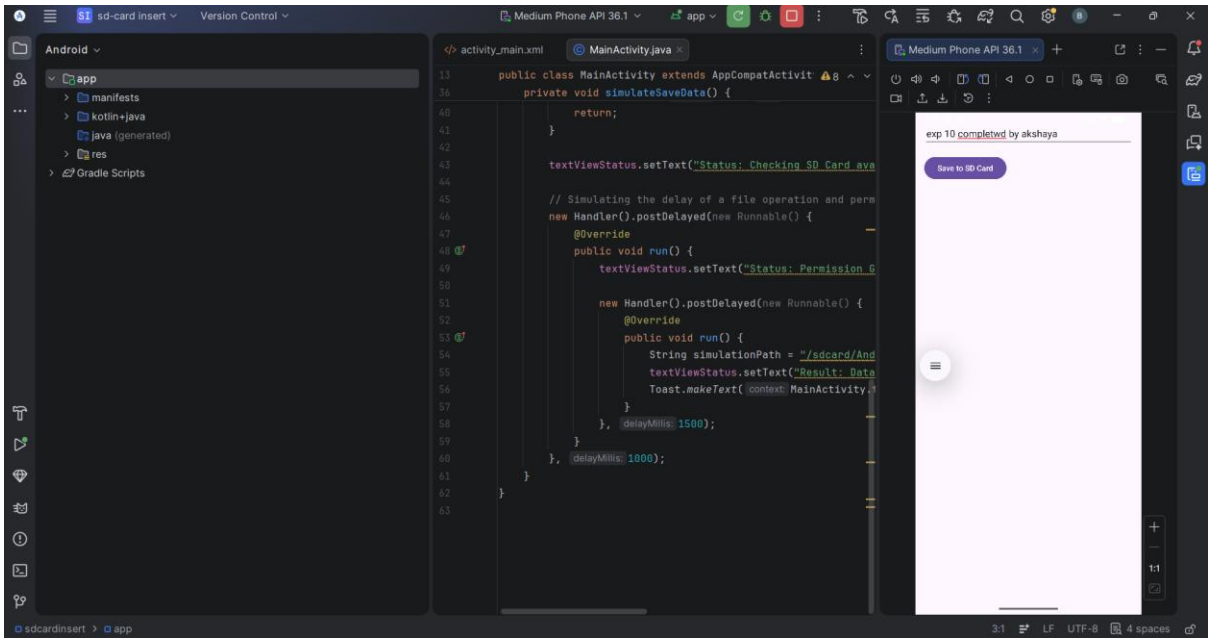
- Data is stored successfully in the SD card
- Runtime permission handling is essential
- External storage path must be handled carefully in newer Android versions

OUTPUT:

- User enters text → Clicks Save
- Message displayed:

Data saved successfully to SD card

- File mydata.txt created in external storage containing user input


sd-card insert
Version Control
Medium Phone API 36.1
app
Android
app
manifests
kotlin+java
java (generated)
res
Gradle Scripts
activity_main.xml
MainActivity.java
public class MainActivity extends AppCompatActivit
private void simulateSaveData() {
return;
}
textViewStatus.setText("Status: Checking SD Card ava
// Simulating the delay of a file operation and perm
new Handler().postDelayed(new Runnable() {
@Override
public void run() {
textViewStatus.setText("Status: Permission G
new Handler().postDelayed(new Runnable() {
@Override
public void run() {
String simulationPath = "/sdcard/And
textViewStatus.setText("Result: Data
Toast.makeText(context: MainActivity.
}
}, delayMillis: 1500);
}
}, delayMillis: 1000);
}
}
Medium Phone API 36.1
exp 10 completwd by akshaya
Save to SD Card
1:1
sdcardinsert
app
3:1
LF
UTF-8
4 spaces

exp 10 completwd by akshaya

Save to SD Card

Result: Data saved successfully to SD card
File: /sdcard/Android/data/com.example.sd_cardinsert/files/mydata.txt
Content: exp 10 completwd by akshaya

Simulation: Data saved to SD Card

RESULT:

An Android application is successfully developed using Android Studio to write data to the SD card with proper permission handling and storage management.